

Problem-Solution Fit

Date	30 September 2026
Team ID	XXXXXX
Project Name	AI StudyBuddy
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience.

1. CUSTOMER SEGMENT(S) [CS] • Primary: School, college and competitive-exam students who study from their own notes, PDFs and text files. • Secondary: Self-learners and working professionals upskilling on their own. • Admins: Platform administrators who manage users and content (role: student / admin).	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited budget: prefers free or low-cost tools. • Mostly phones / low-spec laptops; needs a light web app. • Uploads limited to .txt, .md, .pdf and 5 MB per file. • Very little time before exams; wants zero setup. • Needs internet; concerned about privacy of personal notes.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] • Manual notes / Anki / Quizlet: effective, but slow to create cards. • Generic AI chatbots: powerful, but need manual prompts; no saved library or unified flashcards + quiz + plan. • Tutors / coaching: personalised, but costly and on a fixed schedule. • YouTube / study apps: free, but generic; not based on the student's own material.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Turning long notes into summaries and revision material takes hours (every exam cycle, weekly). • No easy way to self-test; students only re-read (daily). • No structured study plan, so time is wasted deciding what to study (before every exam). • Notes scattered across many files and formats (constant).	9. PROBLEM ROOT / CAUSE [RC] • Study material is voluminous and unstructured. • Converting it into active-recall formats (flashcards, quizzes) is manual and time-consuming. • No personalised, ready-made plan built from the student's own content. • Existing tools are fragmented, do not store outputs, and students lack awareness of techniques like active recall.	7. BEHAVIOR + ITS INTENSITY [BE] • Re-reads and highlights notes: high. • Last-minute cramming near exams: high (seasonal peak). • Ad-hoc chatbot / YouTube searches for explanations: medium-high. • Shares notes with friends: medium. • Creates handwritten flashcards: low.
3. TRIGGERS TO ACT [TR] • Upcoming exam, test or assignment deadline. • Receiving a large chapter, lecture PDF or note set. • Poor mock-test result or an unfinished syllabus. • Seeing classmates use AI tools; peer recommendation.	10. YOUR SOLUTION [SL] • AI StudyBuddy: a secure web app that converts a student's own notes into ready-to-use study material. • Students sign up / log in and upload .txt, .md or .pdf notes (up to 5 MB). • Google Gemini AI automatically generates a summary, flashcards (Q&A), a multiple-choice quiz with answers, and a personalised study plan. • Every output is saved per user in a Material library (with timestamps) to revisit anytime. • Reliable: automatic retries with exponential back-off when the AI service is busy (429 / 503). • Secure: bcrypt-hashed passwords, JWT-protected APIs, and admin-only routes for role-based access. • Stack: Node.js, Express, MongoDB (Mongoose), Multer.	8. CHANNELS OF BEHAVIOR [CH] ONLINE • Web app (AI StudyBuddy client). • WhatsApp / Telegram study groups; Discord, Reddit. • YouTube, Google search, AI chatbots. • College LMS and e-learning portals.
4. EMOTIONS BEFORE / AFTER [EM] • Before: overwhelmed, anxious, procrastinating, unsure what to prioritise. • After: confident, organised and in control, with a clear plan and motivation.		OFFLINE • Classrooms, libraries and hostels. • Peer study groups; teachers and coaching centres. • Textbooks, printed notes and handwritten flashcards.